

## ***House Price Estimation in Yogyakarta Using a Combination of Random Forest and Extreme Gradient Boosting Tree (XGBoost) Models***

Estimasi Harga Rumah di Yogyakarta Menggunakan Kombinasi Model Random Forest dan Extreme Gradient Boosting Tree (XGBoost)

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### ***Abstract***

**Purpose:** This study aims to estimate house prices in Yogyakarta by applying a combination of machine learning models, namely Random Forest and Extreme Gradient Boosting Tree (XGBoost), through ensemble techniques including a meta-model approach.

**Design/methodology/approach:** The study uses a dataset collected from an online property platform, containing features such as land area, building size, number of rooms, and certificate type. The data undergoes preprocessing including normalization, encoding, and handling of missing values. Modeling is conducted using Random Forest and XGBoost individually, then combined using both averaging and a meta-model strategy. Performance is evaluated using MAE, RMSE, and  $R^2$ .

**Findings/result:** The meta-model approach outperforms individual models and the averaging ensemble, achieving the lowest error rates and the highest  $R^2$  value. This confirms the advantage of combining model predictions through a learning-based strategy.

**Originality/value/state of the art:** This study offers a practical ensemble learning framework for house price estimation in urban Indonesia, highlighting the strength of meta-modeling to reduce prediction error and improve generalization, especially in real estate contexts with heterogeneous attributes.

### ***Abstrak***

**Keywords:** Image Classification, SVM, HSV, GLRLM, Corn Seeds.

**Tujuan:** Penelitian ini bertujuan untuk mengestimasi harga rumah di Yogyakarta dengan menerapkan kombinasi model